Highlights

Marginal Conformal Reliability Can Mask Hydroclimatic Failure Modes in Streamflow Uncertainty

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- Conformal intervals under-cover dry and over-cover snow basins.
- Stratification fixes gauged disparity but fails in spatial transfer.
- Variance of log residuals identifies severe undercoverage.
- Average coverage alone is insufficient for reliability.

Marginal Conformal Reliability Can Mask Hydroclimatic Failure Modes in Streamflow Uncertainty

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Abstract

Neural rainfall-runoff models are increasingly paired with conformal prediction for distribution-free streamflow uncertainty intervals, but average reliability can conceal where those intervals fail. Using large-sample experiments on two Catchment Attributes and Meteorology for Large-sample Studies (CAMELS) data sets (United States and Great Britain), spatial-transfer folds, and several uncertainty families, we show that conformal streamflow reliability breaks down along recognizable hydroclimatic gradients. In the United States temporal split, pooled split conformal intervals reach plausible average coverage yet leave a cross-tier coverage spread of 20.97 +/- 1.22 percentage points, systematically under-covering dry basins and over-covering snow basins; CAMELS-GB provides a bounded dataset-native check with an 11.31-point spread. Hydroclimatic stratification removes most of this disparity in the gauged setting (spread 1.53 points) but, under leave-one-region-out spatial transfer, reverses into a dominance violation in which snow basins are covered at only 0.711 basin-pooled coverage by narrower intervals (a seed-42 transfer estimate, reproduced across seeds on the worst-covered folds). The failure is organized by the basin-level variance of signed log residuals computed on the calibration period (Spearman -0.90); in CAMELS-US, an out-of-fold calibration-period screen flags severe per-basin undercoverage with precision 0.74 and recall 0.89. Conformalized quantile regression, a Hopfield-based conformal baseline, a mixture-density model, adaptive conformal inference,

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and static-attribute reweighting each reduce or reshape the disparity but do not make marginal coverage a sufficient criterion. We therefore argue that streamflow uncertainty methods should be evaluated by hydroclimatic conditional reliability and transfer behavior, not by marginal coverage alone.

Keywords: conformal prediction, streamflow uncertainty, hydroclimatic regimes, ungauged basins, CAMELS, neural hydrology

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1. Introduction

Reliable streamflow uncertainty estimates are needed because operational decisions depend not only on a point forecast but also on whether the predictive interval remains credible in the basin and regime where it is used. Neural rainfall-runoff models have become strong point-prediction tools for large-sample hydrology, especially when regional networks learn across many basins rather than only within individual catchments (Kratzert et al., 2018, 2019b,a, 2024; Arsenault et al., 2023). Recent hydrological uncertainty studies have extended these models with probabilistic or conformal outputs (Klotz et al., 2022; Althoff et al., 2021; Li et al., 2024; Auer et al., 2024, 2023). Conformal prediction is attractive in this setting because split conformal calibration can provide distribution-free marginal coverage under exchangeability (Vovk et al., 2005; Shafer and Vovk, 2008; Angelopoulos and Bates, 2023). However, a marginal guarantee over a heterogeneous basin population does not ensure that dry, semi-arid, humid, and snow-dominated regimes are all covered at the intended rate. The practical stakes of this gap are asymmetric: when a nominal 90 percent interval covers a water-limited dry basin well below target while the basin-averaged coverage still looks acceptable, the uncertainty estimate systematically understates hydrological risk in exactly the catchments where drought, water-allocation, and low-flow management decisions are most consequential. Such a failure of predictive reliability is distributed unevenly across recognizable hydroclimatic regimes rather than spread uniformly, so it is invisible to an average-coverage check yet directly relevant to operational use.

The central question of this paper is therefore not whether a new method can solve conditional reliability, but how conformal streamflow reliability fails as a function of hydroclimatic regime, whether stratifying by those regimes helps or harms once predictions are transferred to new regions, and whether the failure is stable enough to anticipate before deployment.

This paper treats hydroclimatic regimes as the first-order reliability structure. The tier design uses aridity and snow fraction because those attributes summarize broad differences in water availability, seasonality, and snowmelt control that can change the distribution of long short-term memory (LSTM) network residuals, and because aridity-snow classifications and streamflow-regime analyses have been shown to carry hydrologically interpretable structure (Knoben et al., 2018; Brunner et al., 2020; Jehn et al., 2020). In both the CAMELS-US (Catchment Attributes and Meteorology for Large-sample Studies, United States) temporal split and the CAMELS-GB data set, pooled global split conformal prediction (Global CP) leaves a large cross-tier coverage spread (quantified in Section 3.1 and Section 3.6), which makes hydroclimatic conditional miscoverage the object of study.

Hydroclimatic stratification is then used as a mitigation and stress-test probe, not as the paper's independent contribution. Hydrologically stratified conformal calibration (HSCC), which is a process-aware Mondrian split conformal baseline rather than a new conformal method, conformalized quantile regression baselines, the Hopfield-based conformal prediction (HopCPT) baseline, the uncertainty mixture of asymmetric Laplacians (UMAL) baseline, Gaussian baselines, the CAMELS-GB data set, and Hydrologic Unit Code level-2 (HUC-2) leave-one-region-out (LORO) transfer all ask the same question from different angles: does a method that looks acceptable marginally still leave hydroclimatic reliability disparities? A named tier also does not automatically mean that residual distributions are exchangeable across regions. Snow-dominated basins in one HUC-2 source pool may not have the same residual variance or seasonal error structure as snow-dominated basins in a held-out region.

Existing work provides the components needed to ask the conditional-reliability question, but not the specific hydrological diagnosis. Probabilistic neural hydrology has developed mixture-density, dropout, and related uncertainty heads for rainfall-runoff models, yet these model-based intervals do not by themselves provide hydroclimatic conditional coverage guarantees (Klotz et al., 2022; Althoff et al., 2021; Li et al., 2024). Hydrological conformal methods such as the Hopfield-based Conformal Prediction for Time series (HopCPT) directly address conformal streamflow uncertainty and temporal information needs, while topology-aware conformal work has begun to connect conformal prediction with stream networks (Auer et al., 2023; Zhang et al., 2025). Distribution-free conformal regression methods such as Conformalized Quantile Regression (CQR) provide strong general baselines (Romano et al., 2019). In parallel, clustered, class-conditional, multivalid, and conditional conformal prediction literatures clarify why group-conditional validity is difficult and why subgroup definitions matter, including the fundamental limits on distribution-free conditional coverage (Barber et al., 2021; Ding et al., 2023; Jung et al., 2022; Bastani et al., 2022; Gibbs et al., 2025). Adaptive conformal inference under distribution shift (Gibbs and Candès, 2021; Zaffran et al., 2022) provides an online coverage-tracking comparator and is included in our cross-family evaluation; its relevance here is diagnostic, because interval width can absorb the cost of conditional reliability.

The hydrological gap is the diagnosis of group-conditional reliability under physically meaningful basin structure. The relevant groups in streamflow are not arbitrary labels; they should reflect process differences that shape runoff generation and model residuals. Prediction in Ungauged Basins (PUB), uncertainty assessment, and regionalization studies have long emphasized that transfer across ungauged or weakly gauged basins depends on catchment similarity and regional process structure (Wagener and Montanari, 2011; Hrachowitz et al., 2013; Razavi and Coulibaly, 2013; Pool et al., 2021). What remains missing is a direct analysis of whether marginal conformal coverage hides hydroclimatic-regime failures, how far alternative uncertainty families reduce that disparity, and where hydroclimatic labels stop carrying enough exchangeability information.

In this work we diagnose this failure mode empirically across uncertainty families and offer a calibration-period indicator, computable for gauged or partially gauged target basins, of where it appears. On the 671-basin CAMELS-US temporal split, pooled split conformal prediction at the 90 percent target leaves a cross-tier coverage spread of 20.97 $\pm$ 1.22 percentage points across five seeds, with dry basins under-covered at 0.746 and snow basins over-covered at 0.955. An analogous dataset-native disparity appears on CAMELS-GB, where Global CP shows a spread of 11.31 $\pm$ 0.05 percentage points and HSCC reduces it by 9.20 $\pm$ 0.43 percentage points, and it is reshaped rather than removed under HUC-2 leave-one-region-out spatial transfer. The empirical contribution is not the observation that runoff residuals are heteroscedastic; it is the quantified size of the hydroclimatic coverage disparity and the finding that a tier calibration that helps in-domain can become non-portable under regional transfer.

Switching the uncertainty family does not by itself dissolve the diagnosis. HSCC, Mondrian-CQR, HopCPT, UMAL, and vanilla Global CQR each reduce but do not eliminate the cross-tier disparity (Section 3.7). Adaptive Conformal Inference brings the spread down to 0.33 percentage points on the available seed-42 run, but only at the cost of extreme-flow interval inflation, with a mean log-width spread of 2.18 across tiers, so marginal coverage does not become a sufficient reliability criterion. Multivalid conformal prediction (Jung et al., 2022) and the Gibbs-Candès conditional conformal framework (Gibbs et al., 2025) are noted as future-work directions and are not claimed to share this diagnosis.

As a practical handle on where the failure will appear, the basin-level variance of signed log residuals computed on the LSTM calibration period, before any test data are consulted, is monotonically related to signed coverage error with Spearman rho -0.901 $\pm$ 0.009 across the five seeds. Section 4.5 turns this signal into a dataset-calibrated screening protocol: in CAMELS-US, a threshold of 0.453 flags severe per-basin undercoverage with out-of-fold precision 0.74 and recall 0.89. This threshold is a validated, calibration-period screening signal for conditional-coverage audit that practitioners can compute before any test outcome is observed; we present it as an actionable audit trigger for gauged or partially gauged basins, with the explicit scope limits that it is a screen rather than a coverage guarantee and is not a substitute for prediction in truly ungauged basins.

2. Methodology

2.1. Datasets and splits

The primary empirical domain is CAMELS-US, which provides catchment attributes and daily meteorological and streamflow records for large-sample hydrology studies (Newman et al., 2015; Addor et al., 2017). The main temporal-split experiments analyze 671 CAMELS-US basins. The record is divided into three consecutive, non-overlapping windows: the LSTM backbone is trained on water years 1980-1990, the conformal calibration set is the held-out NeuralHydrology validation period of water years 1990-2000, and evaluation uses test water years 2000-2014. Each boundary is a water-year cutoff (30 September / 1 October), so the shared endpoint labels mark the split points rather than reused days. Because the calibration scores are computed on the validation period, which is excluded from LSTM training, the split conformal calibration set is independent of the data used to fit the point predictor. The hydroclimatic tier assignment yields 92 dry, 77 semi-arid, 407 humid, and 95 snow basins in the primary CAMELS-US analysis. These tier counts are not used as claims of equal representation; they define the empirical support over which tier-conditional coverage and interval width are evaluated (Table 1).

Table 1: Dataset scopes used in the manuscript. Tier counts are shown for the primary CAMELS-US temporal split; transfer folds are summarized by fold count.

Scope	n	Tier support
CAMELS-US temporal split	671 basins	dry 92; semi-arid 77; humid 407; snow 95
CAMELS-US HUC-2 LORO	18 folds	varies by fold
CAMELS-GB bounded external check	50 primary	dataset-native GB tiers

Roles of these scopes in the manuscript: the CAMELS-US temporal split anchors the primary diagnosis and gauged calibration evidence; HUC-2 LORO supplies the spatial transfer-boundary mapping; CAMELS-GB provides framework portability and a standard uncertainty-quantification scan against dataset-native tiers.

Spatial transfer is evaluated separately from the gauged temporal-split setting. For CAMELS-US, the HUC-2 leave-one-region-out protocol holds out each evaluated region as a target domain. For every fold the LSTM backbone is retrained from scratch on the basins of the remaining regions, and the tier-specific calibration quantiles are estimated from those same remaining regions, so both the point predictor and the conformal calibration exclude the held-out region. This makes the setting a genuine spatial transfer to unseen regions rather than calibration transfer layered on a single globally trained model. The design supports the transfer-boundary claim because it tests whether same-tier residual distributions remain comparable across regions, rather than assuming that a tier label is spatially portable. A bounded external check uses CAMELS-GB, an independently constructed Great Britain large-sample data set (Coxon et al., 2020). The GB analysis intentionally uses dataset-native aridity and snow thresholds, because the method transfers the process-aware partitioning principle, not the numeric US cutoffs.

2.2. Problem formulation

This study treats conformal streamflow uncertainty as a reliability problem over hydrologically meaningful groups. For basin-day observations with discharge Q_i, a point prediction $\hat{Q}_i$, and a calibration set that is exchangeable with the target test set within the scope being claimed, split conformal prediction constructs prediction intervals by calibrating a nonconformity score on held-out residuals (Vovk et al., 2005). The nominal target coverage is 90%, with miscoverage level $\alpha = 0.10$. Because streamflow has a large dynamic range and near-zero flows are common in dry basins, all conformal variants in the main protocol use the log-flow residual score

$$s_i = \left|\log(Q_i + \epsilon) - \log(\hat{Q}_i + \epsilon)\right|,$$

where ϵ is 0.01 millimeters per day. Marginal split conformal prediction estimates one empirical quantile of s_i over the full calibration pool. Hydrologically stratified conformal calibration instead estimates quantiles within process-defined strata, so the object of analysis becomes tier-conditional coverage rather than only average coverage over all basins.

2.3. Temporal dependence and empirical validity

The split conformal coverage statements invoked here hold under exchangeability of calibration and test scores within the scope being claimed. Daily streamflow is strongly autocorrelated and seasonal, basin-days within a basin are clustered, and the calibration and test windows span more than a decade, so basin-days are not strictly exchangeable. We therefore treat split conformal in this work as an empirical post-processing diagnostic rather than as a source of unconditional finite-sample guarantees for daily series, and we restrict any finite-sample coverage statement to the within-tier exchangeability condition. The HUC-2 leave-one-region-out setting is explicitly a covariate- and regional-shift stress test that falls outside the standard split conformal guarantee. Reported cross-seed standard deviations quantify LSTM seed variability. To probe basin-sampling uncertainty, we also report a retained-summary basin-block check that resamples basins and seeds; because daily prediction files are not retained for all non-anchor seeds, this check does not replace a daily or year-block bootstrap.

2.4. Forecast model

All conformal procedures are applied as post-processing layers on a shared LSTM rainfall-runoff backbone implemented with NeuralHydrology (Kratzert et al., 2022). The model follows the large-sample neural hydrology practice of training one regional network across many basins rather than independent basin-specific models (Kratzert et al., 2018, 2019b,a, 2024). The manuscript-facing protocol uses 5 dynamic forcings and 26 static catchment attributes, with hidden size 64 and 30 training epochs. The archived seed-42 CAMELS-US anchor has median basin test Nash–Sutcliffe efficiency (NSE) 0.738 (median across basins, the NeuralHydrology default aggregation), which establishes that the uncertainty analysis is built on a hydrologically competent point predictor rather than on a failed runoff model.

The nonconformity score is the log-flow absolute residual defined above. This score is used consistently for Global CP, HSCC, and conformal baselines when those baselines operate on the shared point-prediction residuals. It reduces the dominance of high-flow magnitudes while retaining sensitivity to low-flow errors, which is important because the principal reliability failures occur across aridity and snow regimes. All reported interval widths are transformed back to discharge units before evaluation, so coverage is assessed on the original streamflow target while calibration is stabilized in log-flow space.

The residual-variance diagnostic uses the signed residual in the same log-flow space, not the logarithm of a variance. For basin b and day t, define $r_{bt} = \log(Q_{bt} + \epsilon) - \log(\hat{Q}_{bt} + \epsilon)$ with the same ϵ used in the nonconformity score. The basin-level diagnostic is $\sigma_b^2 = \mathrm{Var}_{t \in \mathcal{C}_b}(r_{bt})$, computed over the calibration period for gauged temporal-split analyses. In the LORO transfer diagnostic, source-pool variances are computed on calibration-period residuals and held-out target variances on test-period residuals, because the held-out region has no calibration residuals by design.

2.5. Hydrologically stratified conformal calibration as a probe

HSCC is used as a process-aware Mondrian split conformal probe. The procedure first assigns each basin to a hydroclimatic tier using static catchment attributes and then applies the split conformal quantile calculation separately within each tier. In CAMELS-US, snow basins are assigned first when the fraction of precipitation falling as snow is at least 0.4; among the remaining basins, dry basins have aridity above 1.5, semi-arid basins have aridity between 1.0 and 1.5, and humid basins have aridity no greater than 1.0. Here aridity is the CAMELS climatic aridity index, the ratio of mean potential evapotranspiration to mean precipitation, and the snow fraction is the CAMELS attribute giving the fraction of precipitation that falls as snow; both are read directly from the published catchment attributes (Addor et al., 2017). These cut points separate recognizable hydroclimatic regimes – snow-influenced, water-limited arid, transitional semi-arid, and energy-limited humid catchments – and were fixed a priori from hydrological reasoning before any coverage result was inspected, consistent with prior hydrological climate-classification and catchment-clustering work that emphasizes aridity, snow, and seasonality as interpretable regime axes (Knoben et al., 2018; Brunner et al., 2020; Jehn et al., 2020). Their influence is probed by the boundary-sensitivity and random-tier controls reported in the Results rather than tuned to maximize spread reduction. The same principle is used for CAMELS-GB, but with dataset-native thresholds because the GB hydroclimatic distribution collapses under the US aridity and snow cutoffs: the GB primary set assigns montane basins when the snow fraction is at least 0.05, and otherwise splits on aridity into wet (no greater than 0.5), mid (0.5-0.7), and drier (at least 0.7) tiers, yielding 21 wet, 12 mid, 12 drier, and 5 montane basins in the 50-basin primary set, which serves as a bounded external check rather than a full-population GB benchmark. A larger 80-basin set was defined as a potential extension, but every CAMELS-GB result reported in this manuscript uses the 50-basin primary set. The GB montane tier is therefore a dataset-native high-snow/high-elevation tier in this bounded check, not a strict analogue of the CAMELS-US snow regime. This makes the tier design a hydrological covariate-selection problem rather than a fixed numeric threshold transfer. The approach is related to Mondrian and class-conditional conformal prediction, but its role in this manuscript is diagnostic: it tests whether hydroclimatic stratification reduces conditional miscoverage, not whether a new conformal theorem or universal method has been introduced (Ding et al., 2023).

The within-tier calibration rule is straightforward. For a tier g, let $\mathcal{C}_g$ be the calibration examples whose basin belongs to g. HSCC computes the conformal quantile $\hat{q}_g$ from $\{s_i : i \in \mathcal{C}_g\}$ as the $\lceil(|\mathcal{C}_g| + 1)(1 - \alpha)\rceil$-th smallest within-tier score, and returns the log-space interval implied by $\hat{q}_g$ for test points in the same tier. Global split conformal prediction is recovered by replacing $\mathcal{C}_g$ with the full calibration set. The main CAMELS-US protocol analyzes 671 basins and uses 5 seeds for the principal gauged temporal-split evidence, so the manuscript reports cross-seed means and uncertainty for the main coverage-spread claims rather than relying on a single archived run.

The formal guarantee invoked here is intentionally narrow and is the standard split conformal result applied within a stratum: if the calibration and test scores are exchangeable within tier g, the split conformal interval calibrated on $\mathcal{C}_g$ attains finite-sample marginal coverage for future examples from the same tier at the nominal level $1 - \alpha$, up to the standard finite-sample quantile convention (Vovk et al., 2005). This statement justifies only the conditional exchangeability calculation behind a stratum-specific conformal quantile. It does not assert that HSCC solves conditional reliability, that a tier-specific quantile learned in one geographic region remains valid for the same named tier in a held-out region, or that the empirical tier labels are sufficient for universal deployment.

2.6. Diagnostic status of the residual-variance signal

The residual-variance signal used in the transfer-boundary analysis has a precise diagnostic status, which we state as a two-part proposition because a same-period re-description is easily conflated with a non-trivial cross-period claim.

Part (i): same-period re-description (near-algebraic). At a fixed pooled Global CP quantile $\hat{q}$, the coverage of basin b is exactly $\Pr(|R_b| \leq \hat{q})$. If two basins share a common zero-mean standardized residual law up to a scale factor, $R = \tau Z$, then a larger scale (hence larger same-period variance $\mathrm{Var}(R) = \tau^2 \,\mathrm{Var}(Z)$) yields weakly lower coverage, because $\Pr(|\tau_b Z| \leq \hat{q}) \leq \Pr(|\tau_a Z| \leq \hat{q})$ for $\tau_b \geq \tau_a > 0$; more generally the same holds under the threshold-tail ordering $F_{|R_b|}(\hat{q}) \leq F_{|R_a|}(\hat{q})$, that is, whenever basin b places more absolute-score mass beyond $\hat{q}$. In this same-period sense the link between residual dispersion and pooled-quantile undercoverage is a re-description of conformal-score dispersion rather than an independent mechanism.

Part (ii): cross-period predictive claim (falsifiable). The diagnostic actually reported in this manuscript is different and is not an algebraic consequence of Part (i). It ranks basins by the calibration-period quantity $\sigma_b^2 = \mathrm{Var}_{t \in \mathcal{C}_b}(r_{bt})$ on water years 1990-2000 and asks whether that ranking predicts test-period signed coverage error on water years 2000-2014. This requires the additional, non-tautological condition that the basin ordering of residual dispersion is temporally stable in the relevant tail sense between the two periods. The condition is falsifiable: it fails if high-calibration-variance basins become low-variance in the test period, if test-period residual tails change independently of calibration-period σ_b^2, if the LSTM error process drifts across periods, or if a nonzero basin bias μ_b makes the centered variance σ_b^2 understate the absolute-score mass near $\hat{q}$. The non-trivial empirical content of the diagnostic is therefore precisely this cross-period stability, which is what lets the calibration-period variance serve as an a priori screen for future undercoverage. As reported in detail in the Results, the calibration-period screen out-predicts both aridity and mean discharge for severe test-period undercoverage, so the cross-period signal is neither a same-period restatement nor a static-attribute proxy.

The proposition is logically independent of the narrow conformal guarantee above: the diagnostic status of σ_b^2 under Global CP neither strengthens nor weakens the within-stratum exchangeability guarantee. The headline rank correlation between calibration-period σ_b^2 and test-period signed coverage error should therefore be read as a cross-period predictive diagnostic, not as a newly discovered causal mechanism.

2.7. Calibration baselines

The baseline set is designed to separate the main diagnostic questions: whether marginal conformal calibration masks hydroclimatic failures, whether hydrological stratification reduces those failures in the gauged setting, and whether alternative uncertainty methods remove the same conditional-coverage problem. Global split CP uses a single calibration quantile over the pooled calibration set. HSCC uses the same score and split but estimates quantiles within hydroclimatic tiers. The UMAL baseline represents the mixture-density neural hydrology family discussed by Klotz et al. (2022). Because UMAL couples its own point predictor with its uncertainty head, it is a method-system baseline rather than a pure uncertainty-family swap on the shared backbone; its mean test NSE 0.579 in the archived comparison differs from the shared-backbone anchor NSE 0.738, so UMAL tier-spread differences partly reflect point-prediction skill rather than the uncertainty family alone. HopCPT provides a hydrological conformal time-series baseline (Auer et al., 2023) on 574 usable CAMELS-US basins. CQR is represented by two rows: vanilla Global CQR, which applies one pooled conformal correction across 5 seeds, and Mondrian-CQR, which uses tier-aware conformalized quantile regression on 671 basins (Romano et al., 2019). Both CQR rows fit lower and upper conditional quantiles at the 0.05 and 0.95 levels by linear quantile regression on calibration residuals and then apply the conformalized quantile-regression correction of Romano et al. (Romano et al., 2019); Global CQR applies one pooled correction, whereas Mondrian-CQR applies the correction within each hydroclimatic tier.

The comparison protocol is therefore not a single leaderboard or Pareto-frontier claim. Global CP and HSCC isolate the effect of process-aware stratified calibration under identical scores and splits. UMAL tests whether a parametric probabilistic neural model avoids tier-conditional reliability failure. HopCPT tests a hydrological conformal method that uses temporal information differently from split CP, with the explicit caveat that strict filtering leaves 574 of 671 basins. Global CQR and Mondrian-CQR test whether conformalized quantile intervals reduce the same disparity. Results are reported primarily by tier coverage and tier spread, with marginal coverage and interval width retained to show that a method can look good on average while retaining hydroclimatic disparities.

2.8. Transfer-boundary diagnostic

Spatial transfer is therefore treated as a stress test of the diagnosis. In the HUC-2 leave-one-region-out setting, a held-out basin receives the quantile of its hydroclimatic tier estimated from the other regions, and the resulting coverage is interpreted as a test of same-tier-across-region exchangeability. The diagnostic focuses on whether held-out regimes have residual behavior comparable to the source pool, especially through the distribution of σ_b^2, the variance of signed log residuals. This focus is motivated by the observed cross-seed association between σ_b^2 and signed coverage error, with Spearman correlation -0.901 $\pm$ 0.009. The HUC-2 analysis covers 18 held-out regional folds. Its purpose is not to prove transportable coverage under arbitrary covariate shift, a problem that generally requires additional assumptions or reweighting (Tibshirani et al., 2019; Barber et al., 2023), but to test whether the marginal-reliability illusion and hydroclimatic disparity persist when the source and target regions differ. As a rescue stress test, we also estimate weighted conformal quantiles in the LORO setting using diagonal-Gaussian density-ratio weights over source and target aridity and snow fraction only. This static-attribute reweighting uses no target-flow residuals, but because the weights are estimated rather than known likelihood ratios, it is reported as an empirical rescue attempt rather than as a formal weighted-conformal guarantee.

2.9. Evaluation metrics

The primary reliability metric is empirical coverage within each hydroclimatic tier at the nominal target coverage defined in the protocol. Coverage aggregation follows a basin-equal-weight convention: empirical coverage is first computed per basin over its test days, and the reported tier coverage is the unweighted mean of those per-basin coverages. Each basin therefore contributes equally regardless of record length, so large tiers such as humid do not dominate the tier statistic by sheer count of basin-days. The reported marginal (overall) coverage, by contrast, is pooled over all basin-days, and the Global CP calibration quantile is likewise estimated by pooling calibration scores over all basin-days. The main scalar summary is tier coverage spread, computed as the difference between the largest and smallest tier coverages within a method and experimental scope. This metric directly targets the manuscript's diagnosis: a method can achieve plausible marginal coverage while retaining large dry, semi-arid, humid, or snow-regime failures. Mean interval width in millimeters per day is reported alongside coverage to distinguish reliability recovery from uninformative widening. For the residual-variance screening protocol, severe undercoverage is defined as per-basin coverage below 0.85, and the calibration-period σ_b^2 threshold is selected within the target data set or a representative historical basin pool to screen that binary risk label.

Standard uncertainty-quality metrics are included to make the conformal reliability results comparable to hydrological UQ practice. The standard scan reports Winkler score, probability integral transform (PIT) behavior, continuous ranked probability score (CRPS), and reliability deviation in addition to coverage and width. Because a split conformal interval is not itself a full predictive distribution, the PIT and CRPS are computed from a piecewise-linear predictive cumulative distribution reconstructed from each interval: the lower and upper bounds are treated as the $\alpha/2$ and $1 - \alpha/2$ quantiles with a uniform density between them and clipped linear tails outside, so every method is scored on the same reconstructed predictive object. In the CAMELS-US scan, the marginal

CRPS values are 0.932 for Global CP, 0.770 for HSCC, and 0.403 for CQR. These metrics are interpreted as complementary diagnostics rather than substitutes for tier-conditional coverage, because the scientific claim of the paper concerns reliability heterogeneity across hydroclimatic regimes.

2.10. Experimental protocol and pre-specified criteria

The experimental protocol is organized around one empirical claim: marginal reliability can hide hydroclimatic conditional miscoverage. Global CP anchors the marginal conformal case. HSCC tests whether hydrological stratification reduces the spread in a gauged setting. Vanilla Global CQR, Mondrian-CQR, HopCPT, UMAL, Gaussian baselines, and standard UQ scans test whether the finding is an artifact of one uncertainty family. HUC-2 leave-one-region-out and CAMELS-GB test whether the same reliability disparity appears under transfer and external data. The LSTM backbone uses hidden size 64 and is trained for 30 epochs in the main protocol. This structure separates the narrow conformal calculation from the empirical diagnostic evidence.

3. Results

3.1. Global marginal coverage masks hydroclimatic failures

Global split conformal prediction attains an apparently plausible average reliability profile while leaving large hydroclimatic disparities. Across 5 seeds, the Global CP tier coverage spread is 20.97 ± 1.22 percentage points. The direction of the failure is hydrologically structured: dry basins are under-covered, with mean coverage 0.746, while snow basins are over-covered, with mean coverage 0.955. This pattern is the central diagnostic finding of the paper: a pooled conformal quantile can average over regimes that have substantially different residual distributions. Both endpoints of the tier spread use the same basin-equal-weight convention (Section 2.9), so the disparity is not an artifact of mixing the basin-day-pooled marginal aggregation with basin-equal tier statistics; the marginal coverage is reported only to show that average reliability can look acceptable while the conditional spread is large. The pooled marginal coverage of 0.881 in the standard UQ scan is itself below the 90% nominal target, which is a direct empirical reminder that the 1990-2000 calibration window and 2000-2014 test window are not strictly exchangeable. A seed-42 basin-equal-calibration sanity check gives the same conclusion: giving each basin equal total calibration weight changes the Global CP quantile by a factor of only 0.9999 and leaves the tier spread essentially unchanged (19.86 versus 19.87 percentage points), with dry coverage still near 0.750 and snow coverage near 0.949.

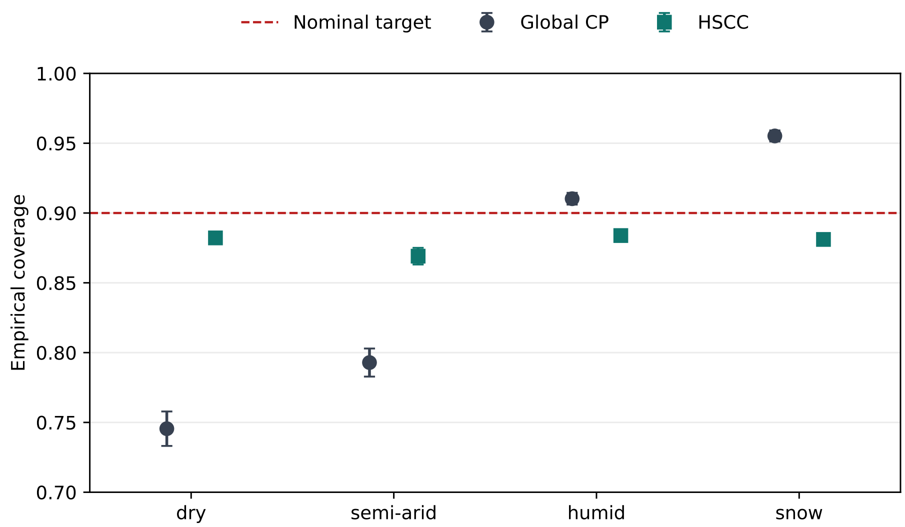

Figure 1: Tier-conditional coverage under Global CP (grey circles) and HSCC (teal squares) in the CAMELS-US temporal-split setting (n = 5 seeds), shown as point estimates with cross-seed standard-deviation error bars. The dashed line marks the nominal 0.90 target. A point-range display is used instead of bars so that coverage is read against the target rather than against an arbitrary zero baseline.

The ablation suite, evaluated on the seed-42 anchor run as a single-seed robustness probe (baseline spread reduction 18.86 percentage points), supports the interpretation that the tier structure is not an arbitrary partition. Random-tier permutation places the observed spread reduction far into the upper tail of the random baseline (z score 10.91); because the null is only 50 permutations deep, we report the exact permutation result rather than the parametric tail: no random-tier partition reached the observed reduction, an empirical one-sided p below 0.02. Boundary sensitivity is a post hoc stress test of the a priori cut points, not a tuning procedure: HSCC spread reduction remains between 12.92 and 21.60 percentage points across all 15 evaluated threshold combinations, and 12 of them exceed the stricter 15-point reduction screen — short of the pre-registered 13-of-15 target, which this stress test does not meet. The 3 combinations that miss that screen all use the most restrictive snow cutoff of 0.50, which labels the fewest basins as snow (73 versus 123 snow basins at the loose 0.30 cutoff), rather than a loose snow cutoff; the single worst case (12.92 percentage points) pairs that cutoff with the widest aridity band, and one of the three keeps the baseline aridity boundaries. The failures therefore occur at the most snow-sparse edge of the sensitivity grid, and even there the spread reduction stays above 12.9 points. The leave-one-regime stress test passes in all four held-out-regime cases. These checks do not prove that the selected tier boundaries are unique, but they make the diagnosis more robust than a single post hoc split.

The conditional-coverage failure is also not an artifact of an undersized or undertrained backbone. Because the diagnosis rests on the residual structure of the LSTM, and undertraining could systematically inflate residual variance and exaggerate tier differences, we retrained the CAMELS-US temporal-split model on the same seed-42 split with a 256-unit hidden state for 50 epochs, a four-times-wider and longer-trained network than the 64-unit, 30-epoch anchor. The larger backbone attains comparable point skill, with median basin test NSE 0.695 against 0.738 for the anchor at the same seed, and reproduces the failure essentially unchanged: the Global CP tier coverage spread is 19.14 percentage points, against 19.86 for the anchor at the same seed, with the same structure of dry under-coverage (0.762) and snow over-coverage (0.953). Quadrupling capacity and extending training therefore does not remove the disparity; if anything the additional capacity does not improve point skill here, which is itself evidence that the anchor is not underfit. The larger backbone reused the anchor's optimizer, learning-rate schedule, and deliberately short single-decade

(1980–1990) training window without re-tuning for the added capacity, so its 0.695 median NSE is best read as a conservative lower bound on the wider model's achievable point skill rather than a capacity ceiling; the robustness conclusion depends only on the failure structure being reproduced. This is a single-seed check rather than a full multi-seed replication.

3.2. HSCC as a gauged mitigation probe

HSCC sharply reduces cross-tier reliability disparities in the gauged temporal-split setting, so it is a useful mitigation probe for the diagnosis. The HSCC tier spread is 1.53 ± 0.51 percentage points, corresponding to a Global CP to HSCC spread reduction of 19.44 percentage points with bootstrap interval [18.46, 20.65]. A retained-summary basin-block and seed bootstrap gives the same qualitative separation: Global CP spread 21.09 percentage points with 95 percent interval [16.23, 26.09], compared with HSCC spread 3.24 with interval [0.90, 6.53]. The cross-seed HSCC coverages are 0.882 for dry, 0.869 for semi-arid, 0.884 for humid, and 0.881 for snow basins.

The semi-arid tier requires explicit disclosure. Its mean coverage is stable but remains below 0.88, so the manuscript does not describe HSCC as restoring conditional reliability. The defensible claim is narrower: HSCC shows that hydroclimatic stratification can reduce cross-tier spread in a gauged setting while keeping remaining conditional offsets visible.

The tier means also conceal substantial per-basin heterogeneity, which is why the basin-equal-weight distribution is reported alongside the tier mean (Table 2). Even when HSCC moves a tier mean close to the target, the worst-decile (10th-percentile) per-basin coverage remains far below nominal: under HSCC the dry tier has 10th-percentile coverage 0.656 and the snow tier 0.598. Stratified recalibration therefore corrects the tier average but does not eliminate per-basin coverage heterogeneity, consistent with the conditional-miscoverage framing. These per-basin distributions are reported for the seed-42 run as a representative seed; the cross-seed tier means in Figure 1 confirm the same qualitative ordering.

Table 2: Basin-level coverage distribution (seed-42 CAMELS-US) for Global CP and HSCC, summarizing the spread of per-basin coverage within each tier rather than only the tier mean. Median is the typical basin; p10 is the worst-decile basin; the last column is the fraction of basins below 0.85 coverage. HSCC lifts the dry-tier distribution but leaves a wide lower tail, and it tightens the over-covered snow tier at the cost of a heavier lower tail.

Tier	Method	Median basin cov.	Worst-10% (p10) cov.	Frac. basins < 0.85
dry	Global CP	0.796	0.406	0.609
dry	HSCC	0.930	0.656	0.283
semi-arid	Global CP	0.833	0.562	0.519
semi-arid	HSCC	0.910	0.689	0.273
humid	Global CP	0.978	0.703	0.206
humid	HSCC	0.959	0.644	0.260
snow	Global CP	0.995	0.852	0.084
snow	HSCC	0.960	0.598	0.242

3.3. Variance of signed log residuals explains coverage error as correlation

The residual-variance mechanism is reported as a correlation, not as a causal proof. Across seeds, the Spearman association between variance of signed log residuals and signed coverage error is -0.901 ± 0.009. The sign is consistent with the main diagnosis: basins with larger residual variance tend to be under-covered by a pooled conformal quantile, while basins with smaller residual variance tend to be over-covered (Figure 2). This association gives a compact empirical explanation for why aridity and snow regime can organize coverage error. We read it as a re-description of the calibration-score distribution rather than as an independent physical mechanism: the variance of signed log residuals is, up to sign, the dispersion of the conformal score itself, so under a single pooled quantile larger-variance basins are mechanically under-covered. Consistent with this scoping, the two physical error properties we tested — residual autocorrelation (AR1) and heteroscedasticity — do not order the magnitude of per-basin miscoverage in the seed-42 diagnostics (Spearman 0.03 and -0.04 respectively against |coverage - 0.90|), so the pre-registered mechanism gate that required either to reach a rank correlation of 0.4 with miscoverage was not met. The operative quantity is therefore the spread of the conformal score, not a specific physical error mode.

The same-period and cross-period versions of this association carry different evidentiary status. At a fixed pooled quantile within a single period, the ordering of basins by residual dispersion and by undercoverage is a by-construction redescription, formalized as the residual-dispersion proposition in the Methodology (Section 2.6). The headline cross-seed correlation is the stronger, cross-period claim, because it pairs calibration-period variance of signed log residuals (water years 1990-2000) with test-period signed coverage error (water years 2000-2014), so its non-tautological content is that the calibration-period dispersion ranking stays predictive of test-period undercoverage rather than being reshuffled across the decade. That cross-period predictive value is also not merely a static-attribute proxy: used as an a priori screen for severe test-period undercoverage, calibration-period variance of signed log residuals attains an area under the ROC curve of 0.939, against 0.730 for aridity and 0.784 for mean discharge, an improvement of 0.209 and 0.154 in rank-screening AUC, with out-of-fold balanced accuracy 0.881 over the 670 of 671 basins with a defined mean-flow signature (one CAMELS-US basin lacks the mean-flow attribute used to construct the naive baselines and is dropped so that every predictor, including the variance of signed log residuals, is scored on the identical basin set). The diagnostic therefore carries cross-period information beyond catchment climate and mean flow, which is the part of the signal that is not a same-period algebraic restatement.

Calibrating this signal as a screening protocol gives a useful but deliberately scoped diagnostic. For CAMELS-US Global CP, a threshold variance of signed log residuals >= 0.453, selected and scored in sample on the pooled 3,355 basin-seed rows (671 basins × 5 seeds), flags severe per-basin undercoverage (coverage below 0.85) with precision 0.760, recall 0.889, specificity 0.890, and balanced accuracy 0.889 (TP/FP/TN/FN = 841/266/2,143/105; positive rate 0.282, so precision is evaluated against a non-balanced base rate). Because those pooled rows treat the 671 basins under five seeds as independent observations and choose the threshold in sample, they are an optimistic reference. Re-estimated at the basin level on the 671 independent basins (seed 42) with an out-of-fold threshold (5-fold cross-validation), the same screen retains precision 0.74, recall 0.89, and balanced accuracy 0.885 — essentially unchanged from the in-sample basin-level value (balanced accuracy 0.896), so the screen's usefulness does not rest on the pooled, in-sample estimate. The bounded CAMELS-GB check uses a different calibrated threshold, 0.158, and a different operating profile: precision 0.769, recall 0.857, specificity 0.922, and balanced accuracy 0.889 over 150 basin-seed rows (50 basins × 3 seeds; TP/FP/TN/FN = 30/9/106/5). Although the two balanced accuracies both round to 0.889 at three decimals, the underlying sensitivities, specificities, sample sizes, and count tables differ; the useful object is therefore the calibration protocol, not the US numeric cutoff as a universal constant.

The controlled synthetic perturbation check did not convert that association into a causal claim. The AR1 perturbation test has p value 0.184, and the HSCC coverage perturbation test has p value 0.968, even though the same perturbation did significantly degrade point skill (NSE perturbation p value 0.009); the manipulation was therefore effective on the forecast without moving conditional coverage in the hypothesized direction. These null results are retained as a scope limitation. They argue against writing the mechanism section as proof that one residual statistic causes the observed coverage failures, while still supporting its use as a boundary diagnostic for residual-distribution mismatch.

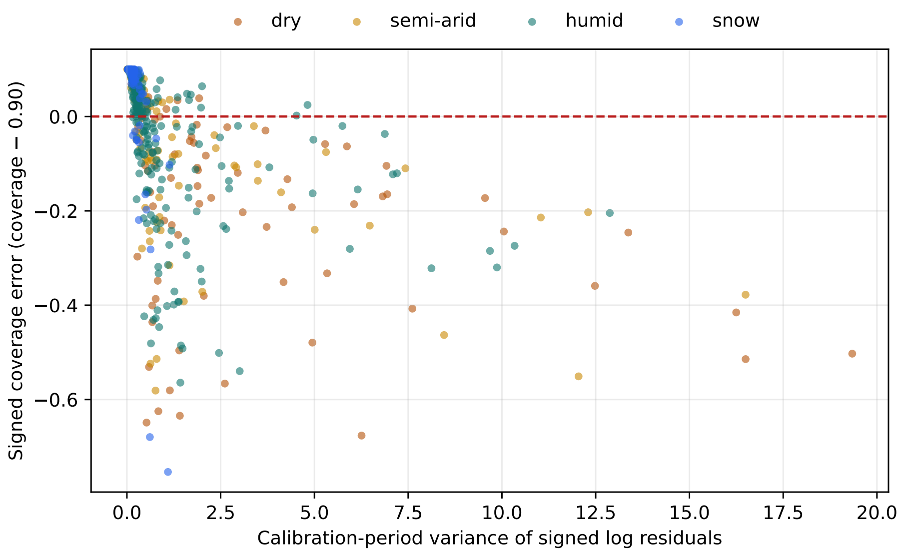

Figure 2: Per-basin calibration-period variance of signed log residuals versus signed coverage error (empirical coverage minus the 90% nominal target) under Global CP, for the seed-42 CAMELS-US run, colored by hydroclimatic tier. Basins with larger variance of signed log residuals are systematically under-covered; the cross-seed Spearman correlation is reported in the text. The dashed line marks zero coverage error.

3.4. Transfer-boundary mapping under HUC-2 LORO

The HUC-2 leave-one-region-out (LORO) analysis changes the interpretation of hydroclimatic stratification from mitigation to stress test. Across 18 evaluated held-out regional folds, the fold-averaged Global CP tier spread is 22.45 percentage points, and the fold-averaged HSCC tier spread is 12.36 percentage points; here "fold-averaged" is the mean across the 18 folds of each fold's own tier spread, distinct from the basin-pooled convention (all held-out basins of a tier pooled; used in Figure 3 and the weighted-conformal comparison below). Stratification reduces average disparity under spatial transfer, but the residual spread remains large and therefore supports the conditional-miscoverage diagnosis rather than a transfer solution.

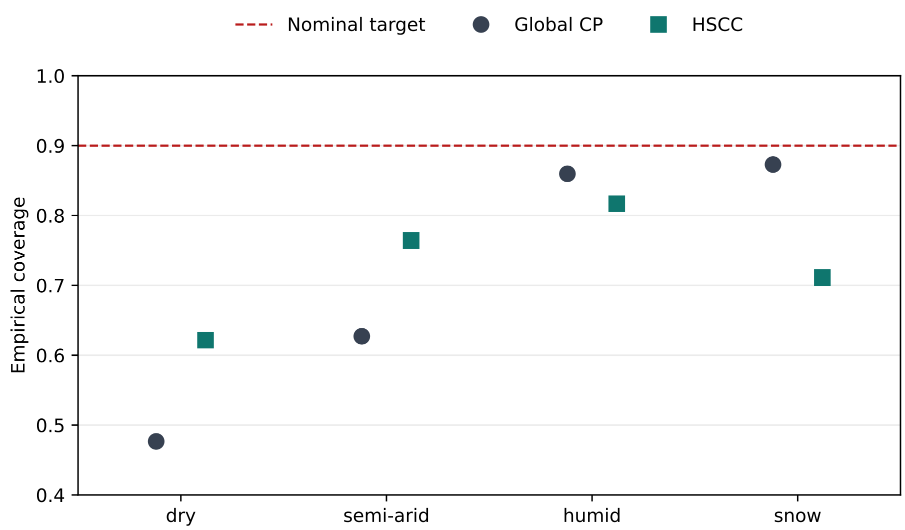

Figure 3: Tier-conditional coverage under HUC-2 leave-one-region-out transfer, pooled basin-equal over all held-out basins of each tier across the 18 folds (snow n = 95), with Global CP shown as grey circles and HSCC as teal squares. This basin-pooled view uses a different aggregation convention from the region-tier heatmap: basin-rich regions carry more weight, and the all-snow pooled summary retains the two one-basin snow cells that are blanked in the heatmap for visual stability. The dashed line marks the nominal 0.90 target; the snow-tier drop under HSCC is the main dominance violation.

The most important boundary case is the snow tier. Under LORO transfer, Global CP covers snow basins at 0.873, while HSCC covers them at 0.711. Unlike the five-seed gauged numbers, these transfer coverages are reported on the seed-42 anchor folds and are therefore a single-seed transfer estimate (Section 4.4). The headline number is single-seed, but the violation itself is not an artifact of one training seed: re-running the four worst-covered held-out folds with three seeds (42, 137, and 2024) reproduces it with small cross-seed scatter, giving held-out snow HSCC coverage 0.450 ± 0.017 for HUC-18 against Global CP 0.646, and 0.633 ± 0.034 for HUC-14 against 0.799. The snow violation therefore behaves as a systematic within-tier-across-region exchangeability boundary rather than seed noise, even though a full eighteen-fold multi-seed transfer is left to future work and the precise pooled value should be read as a seed-42 point estimate. This is a dominance violation because the stratified interval is also narrower: snow-tier mean width is 4.49 millimeters per day for HSCC compared with 7.52 millimeters per day for Global CP. The six snow-bearing heatmap cells with at least 3 held-out snow basins give the same conclusion: their basin-weighted HSCC snow coverage is 0.707, while their unweighted region-cell mean is 0.684. The attempted cookbook classifier reinforces the boundary interpretation. Its leave-one-HUC-out random forest accuracy is 0.344, below the tier-only baseline of 0.469, with paired-bootstrap p value 0.926. A later fallback-rule audit also downgraded fallback logic to exploratory diagnostic status, so the manuscript treats these outputs as warning signals rather than operational prescriptions. Static-attribute weighted conformal calibration does not rescue the boundary: it improves LORO HSCC mean coverage from 0.728 to 0.759 and reduces the basin-pooled tier spread from the baseline HSCC value 19.52 to 15.83 percentage points, but the mean interval width increases by a factor of 28.4. This 19.52-point baseline is the basin-pooled HSCC tier spread (pooled humid minus dry coverage in Figure 3), larger than the 12.36-point fold-averaged spread above because of the aggregation convention, not a discrepancy. The region-by-tier structure of this transfer behavior is summarized in Figure 4, which makes explicit where calibration transfer succeeds, under-covers, or over-covers across held-out HUC-2 regions.

The snow dominance violation is not uniform across regions, and this is where the variance of signed log residuals and the transfer map connect. In Figure 4 the HSCC snow column ranges from 0.47 (HUC-18) up to 0.87 (HUC-13), so the pooled snow mean is dragged down by a subset of regions rather than by a uniform failure. Using the same anchor LORO folds as the heatmap, Figure 5 pairs each snow-bearing region with its source-to-target shift in variance of signed log residuals. The Spearman association between this shift and held-out snow coverage is -0.43 ($p = 0.40$). We use the magnitude of the residual-variance shift here because it is the quantity the calibration-period variance-of-signed-log-residuals mechanism in Figure 2 directly predicts; the two-sample Kolmogorov-Smirnov distance between the same source and target residual-variance distributions, which is the distributional-distance form proposed as a screening test in Section 4.5, is also recorded in the archived per-region data but is even noisier at this region count (its rank association with held-out snow coverage is -0.20, reported against the same coverage axis as the residual-variance shift). With only 6 snow-bearing held-out regions, this panel is therefore a hypothesis-generating consistency check rather than a statistically conclusive predictor of regional snow failure.

The point-skill coloring in Figure 5 supports the narrower claim that the snow boundary is not simply the lowest-skill folds failing. Across the same 6 snow-bearing held-out regions, fold-level point skill does not positively predict snow coverage; the Spearman association between held-out fold median NSE and HSCC snow coverage is -0.26. The HUC-13/HUC-14 dissociation makes this concrete: the snow region with the weakest point skill (HUC-13, fold median NSE 0.02) retains the highest snow coverage (0.87), while the snow region with the strongest point skill (HUC-14, NSE 0.54) is among the most severely under-covered (0.59). The residual-variance and point-skill diagnostics therefore narrow the interpretation: the transfer-boundary failure is observed directly in coverage, while the residual-shift panel suggests a plausible mechanism that requires larger regional replication before it can become a predictive rule.

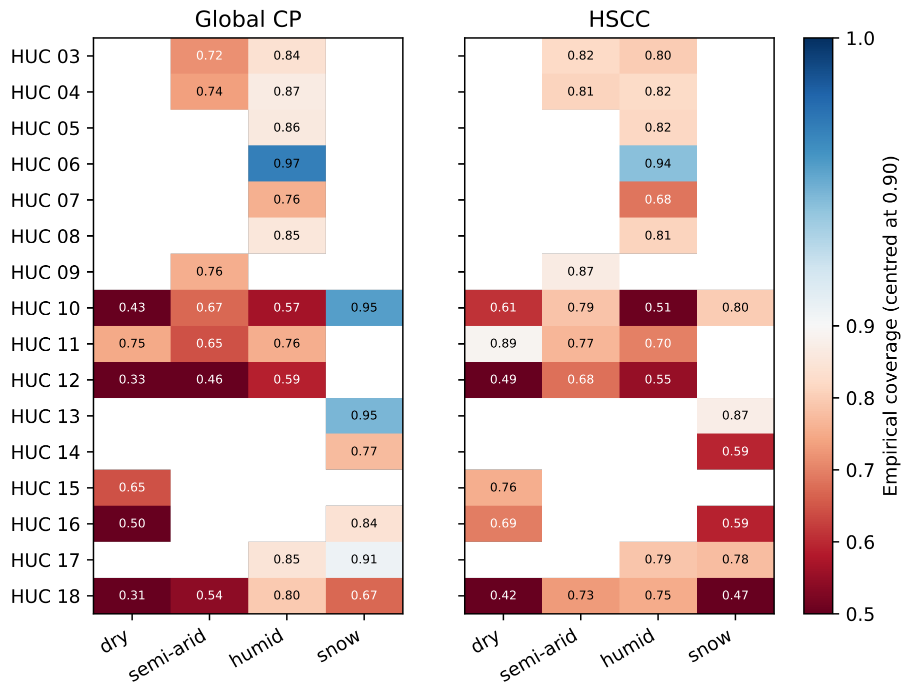

Figure 4: HUC-2 leave-one-region-out coverage by held-out region (rows) and hydroclimatic tier (columns), for pooled Global CP (left) and HSCC (right). Color is empirical coverage centred on the 90% nominal target (blue over-covered, red under-covered); the numeric value is printed in every cell so the panels stay legible in grayscale and under color-vision deficiency. Blank cells have fewer than three held-out basins, the minimum for tier aggregation, so single-basin cells are not shown as stable estimates. The HSCC snow column shows the transfer-boundary failure where stratification under-covers relative to Global CP.

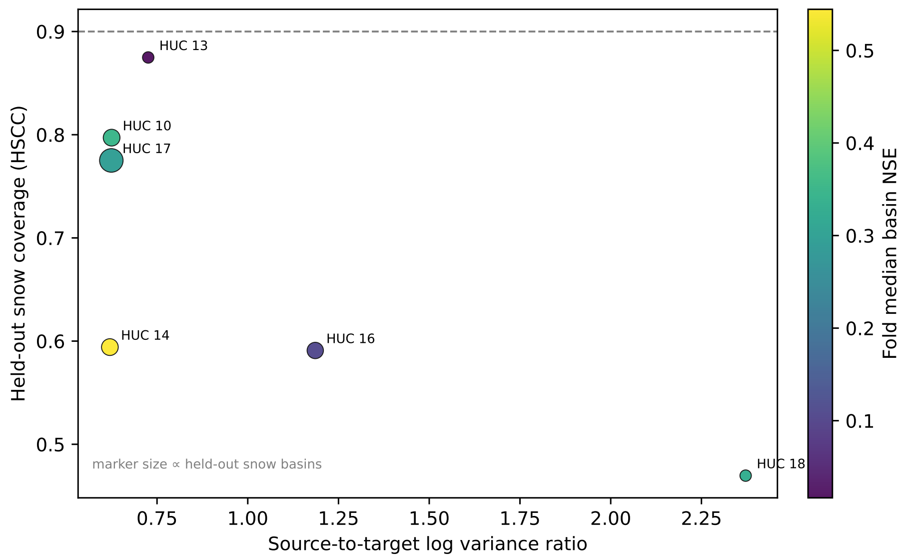

Figure 5: Residual-variance mechanism versus the snow transfer boundary across the six held-out HUC-2 regions with at least three snow basins (same anchor LORO folds as the region-tier heatmap). Horizontal axis: source-to-target log ratio of residual-variance magnitude, log(median held-out-region variance of signed log residuals / median source-pool variance); larger values mean the held-out region's snow residuals are more dispersed than the calibration pool the HSCC snow quantile is built from. Vertical axis: held-out snow coverage under HSCC, with the 90% nominal target marked. Markers are colored by the fold's median basin test NSE, so residual-variance ordering can be read separately from point skill. Source residuals are calibration-period and target residuals test-period, an asymmetry inherent to leave-one-region-out transfer; regions with a larger residual-variance shift are generally under-covered more severely.

The point-prediction skill of the eighteen retrained transfer models is reported so that the snow result is not read as a calibration failure where it is in part a point-skill failure. Each fold retrains the LSTM from scratch on the same protocol (hidden size 64, 30 epochs) and is evaluated on its held-out region. The held-out median basin NSE varies widely across folds, from 0.02 (HUC-13) to 0.70 (HUC-05), with a cross-fold median of 0.44 (Figure 6). No fold has a negative median NSE, but 4 of 18 folds fall below 0.3, indicating weak point transfer into a few mountainous and arid regions. This is a necessary caveat for the lowest-skill arid folds, where weak point transfer and calibration-transfer failure overlap and cannot be fully separated. It does not, however, explain the snow boundary as a whole: as shown above, point skill does not order snow coverage across the held-out snow regions, the weakest-skill snow region is in fact the best covered, and the most severe snow under-coverage is not confined to the lowest-skill regions. The diagnosis that pooled conformal calibration cannot guarantee conditional reliability under spatial transfer therefore holds, and the snow violation is best read as a calibration-transfer effect that is reinforced, but not produced, by point-skill loss in the few lowest-skill folds.

3.5. Random-tier control

The random-tier control tests whether the observed spread reduction can be reproduced by arbitrary partitions. This control is evaluated in the HUC-2 leave-one-region-out setting, so the reference is the LORO spread reduction (the 22.45 to 12.36 percentage-point reduction of Section 3.4), not the larger gauged reduction of Section 3.1. Physical hydroclimatic tiers reduce the LORO spread by 10.09 percentage points on average, whereas random tiers of the same sizes average -0.33 percentage points. Taking the held-out fold as the unit of analysis (the 18 folds, of which 13 contain more than one tier and can discriminate the partitions), a paired test of physical versus mean-random fold reduction gives $t = 4.39$ (one-sided $p = 2.0\text{e-}04$), with a nonparametric Wilcoxon one-sided $p = 9.4\text{e-}04$ and paired effect size $d_z = 1.03$; we report this fold-level test rather than pooling the 360 within-fold random draws, which are not independent.

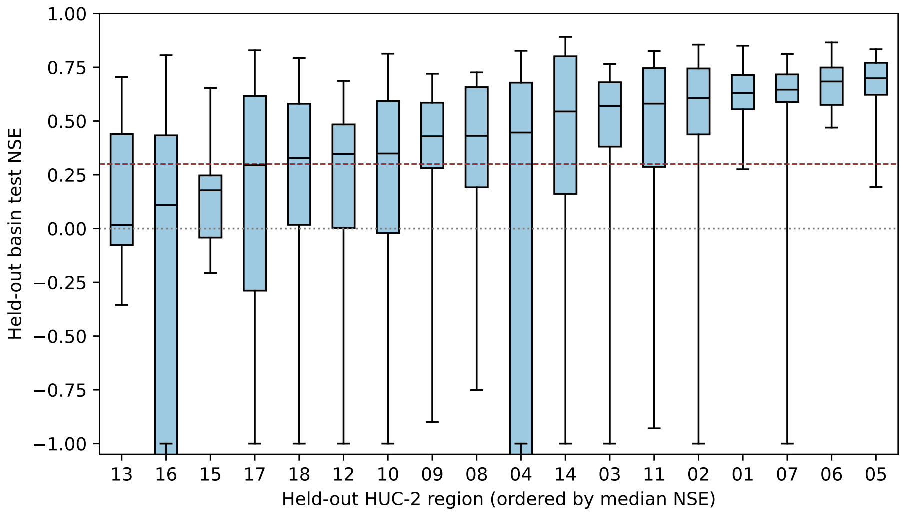

Figure 6: Distribution of held-out basin test NSE for each of the 18 HUC-2 leave-one-region-out folds, ordered by median, computed from the same retrained models whose transfer coverage is reported above. Whiskers are clipped at NSE -1 for display; a few arid and flashy basins have large negative NSE. Most folds retain useful point skill, but several mountainous and arid regions transfer poorly — the point-skill caveat to the snow boundary result.

This result supports the hydrological relevance of the partition, while the LORO snow result shows that hydrological relevance is not enough to guarantee spatial transfer.

3.6. CAMELS-GB bounded external check

CAMELS-GB provides a bounded external test of the diagnostic under dataset-native tier definitions. Across 3 seeds, Global CP has tier spread 11.31 ± 0.05 percentage points, showing an analogous marginal-reliability illusion outside CAMELS-US. The cross-seed standard deviation is small because tier assignment is fixed by seed-invariant static catchment attributes and the GB LSTM is highly stable across seeds (NSE 0.857 ± 0.001), so only the within-tier calibration quantiles vary across seeds while the tier composition and point predictions are nearly fixed. HSCC reduces the GB spread to 2.11 ± 0.47 percentage points. The corresponding LSTM NSE is 0.857. A bounded CAMELS-GB CQR check is more conservative: Global CQR has spread 12.63 percentage points and Mondrian-CQR has spread 5.92 percentage points on the 50-basin GB set. The tier-resolved GB coverages are shown in Figure 7. These results support the diagnosis, not a complete cross-dataset frontier claim or a strict replication of the CAMELS-US snow-regime result.

3.7. Cross-family diagnostic comparison

The baseline comparison (Table 3) shows that the main diagnosis is not resolved by simply switching uncertainty families. Because the rows differ in basin count, seed count, and point-prediction skill, this comparison is read as a cross-family diagnostic scan under partially harmonized protocols rather than as a single leaderboard. UMAL has tier spread 9.88 percentage points and remains below nominal marginal coverage. HopCPT has tier spread 2.48 percentage points and marginal coverage 0.931, but it uses 574 basins after strict filtering. Vanilla Global CQR has marginal coverage 0.8925 ± 0.0008 with tier spread 4.75 ± 0.56 percentage points. Mondrian-CQR has tier spread 2.38 percentage points, marginal coverage 0.890, and mean width 2.00 millimeters per day. Adaptive

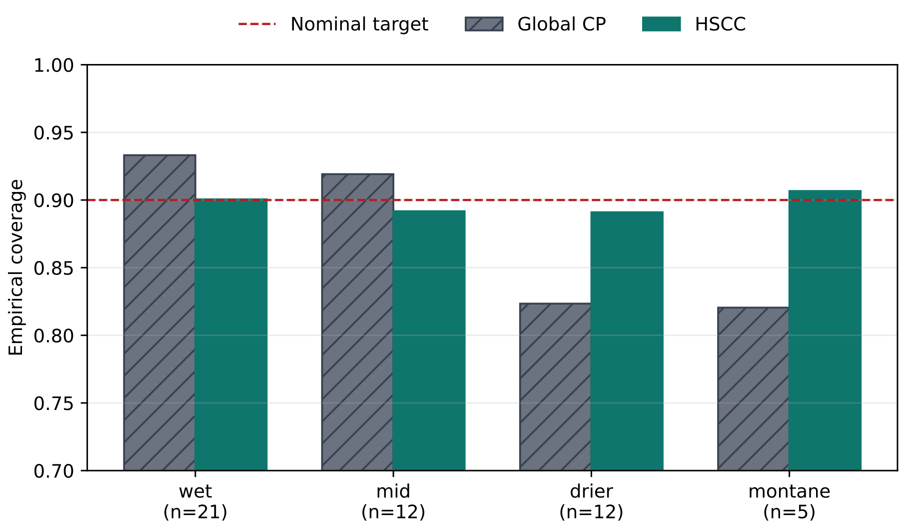

Figure 7: CAMELS-GB tier-conditional coverage under pooled Global CP (grey hatched) and HSCC (teal) on the dataset-native primary tiers (basin counts shown under each tier). The dashed line marks the 90% nominal target. Global CP over-covers the wet and mid tiers and under-covers both the drier and montane tiers (montane, the GB high-snow/high-elevation analogue, is in fact the most under-covered and is the spread-defining minimum tier), and HSCC compresses the tiers toward the target in this bounded external check.

Conformal Inference (Gibbs and Candès, 2021) is the most aggressive comparison because it is designed specifically to chase a target coverage online: on the same seed-42 LSTM checkpoint and the same 671 basins, ACI attains marginal coverage 0.8998 and tier coverage spread 0.33 percentage points. That narrow coverage spread is informative, but it does not by itself close the diagnosis because the log-flow score makes interval-width behavior most visible on extreme-flow days. Because the conformal interval is multiplicative in discharge units, with upper bound scaling as $\widehat{Q}\exp(q_t)$, the typical interval remains hydrologically plausible while its linear-scale upper tail is extreme. The per-basin median width is 2.05, 2.58, 3.87, and 2.05 millimeters per day for the dry, semi-arid, humid, and snow tiers, but as ACI drives the online quantile q_t upward to chase coverage, a few high-flow days inflate the 95th-percentile per-basin width to roughly 702 thousand millimeters per day in the humid tier. The natural-scale summary that is robust to this multiplicative tail is the across-tier mean log-width spread of 2.18 (a factor of approximately 8.8 between the widest and narrowest tier interval at the 90 percent level). Under this log-score implementation, without an upper bound on the online conformal quantile q_t (gamma = 0.005), ACI leaves extreme-flow widths needing additional treatment rather than by itself resolving the diagnosis (Figure 8). This width blow-up is an artifact of the unbounded multiplicative back-transform rather than an intrinsic limitation of online conformal calibration: re-running ACI with a physically motivated cap that limits each interval's upper endpoint to 2 times the basin's maximum calibration-period observed flow binds on only 1.0 percent of basin-days, yet it collapses the 95th-percentile humid-tier width from roughly 702 thousand to 14.8 millimeters per day while leaving marginal coverage (0.900) and tier coverage spread (0.33 percentage points) essentially unchanged; the per-tier median widths shift only marginally, because the cap acts almost entirely on the extreme upper tail rather than on the bulk of the interval distribution. With the cap, ACI attains a near-zero tier spread at physically plausible widths, but it does so by per-day online width adaptation rather than by resolving the conditional-reliability question, so a bounded ACI sharpens rather than overturns the conclusion that a single marginal calibration is not sufficient. These rows reduce or reshape the disparity, but they do not make marginal coverage a sufficient criterion.

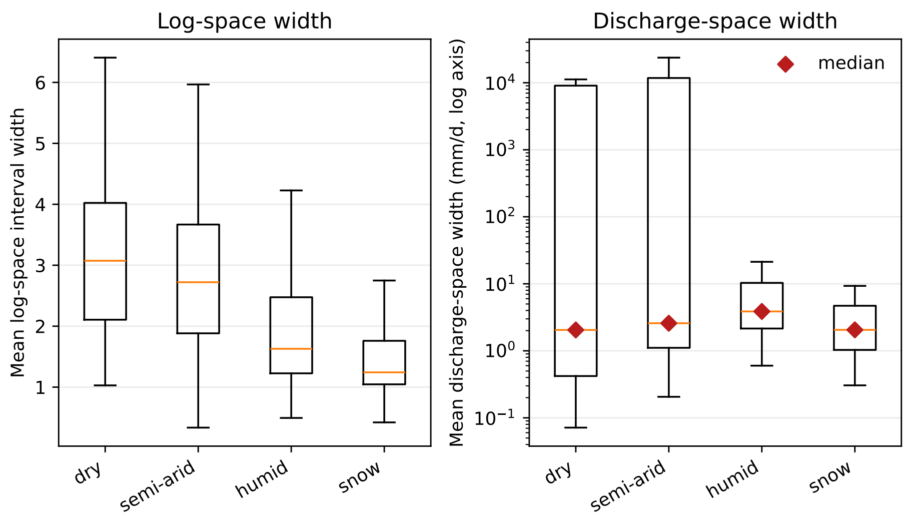

Figure 8: Per-basin ACI interval width by tier on the seed-42 CAMELS-US run. Left: mean log-space width, well behaved across tiers. Right: mean discharge-space width (log axis), heavy-tailed because the log-space interval is multiplicative ($\hat{Q}\exp(q_t)$); a few extreme-flow days drive the upper tail to physically implausible values even though the per-basin median width (diamonds) stays near a few millimeters per day. The log-score ACI implementation thus controls coverage while leaving extreme-flow width to be managed.

Table 3: Cross-family diagnostic comparison of uncertainty methods. Global CP, HSCC, and Global CQR use five seeds; UMAL and HopCPT use three seeds; ACI and Mondrian-CQR use the seed=42 LSTM predictions only, so rows are diagnostic checks rather than equal-depth replications. The Tier-spread column is the cross-seed mean where multiple seeds are available, whereas the marginal Coverage column for Global CP, HSCC, and UMAL is reported at the anchor seed (seed 42) from the standard-UQ scan; single-seed rows therefore carry no cross-seed uncertainty and should not be read against the multi-seed rows as a leaderboard. n is the basin count. CRPS is in millimeters per day. For the conformal interval methods it is computed on the piecewise-linear interval predictive distribution; for UMAL it is computed on its native sample-based predictive distribution, so the UMAL value reflects a sharper density and is not directly comparable cell-for-cell. CRPS is left blank for HopCPT, Global CQR, and ACI, whose archived outputs do not expose a predictive object on the same target.

Method	Role	n	Seeds	Tier spread (pp)	Coverage	CRPS
Global CP	pooled split conformal baseline	671	5	20.97	0.881	0.932
HSCC	stratified mitigation probe	671	5	1.53	0.882	0.770
UMAL	probabilistic neural hydrology baseline	671	3	9.88	0.729	0.406
HopCPT	hydrological conformal time-series baseline	574	3	2.48	0.931	not reported
Global CQR	pooled conformal regression replication	671	5	4.75	0.893	not reported
Mondrian-CQR	tier-aware conformal regression replication	671	1	2.38	0.890	0.403
ACI (Gibbs 2021)	online conformal baseline	671	1	0.33	0.900	not reported

3.8. Standard UQ metrics

The standard UQ scan gives a complementary view of the same methods. CAMELS-US marginal CRPS is 0.932 for Global CP, 0.770 for HSCC, and 0.403 for CQR, all in millimeters per day on the piecewise-linear interval predictive distribution. Evaluated on its native sample-based predictive distribution, UMAL attains a lower marginal CRPS of 0.406 ± 0.011 millimeters per day across three seeds, as expected for a continuous density relative to a two-knot interval CDF. This is the point of the comparison rather than a counterexample to it: UMAL is the sharpest distributional model in the table yet still shows a tier coverage spread of 9.88 percentage points, so a strong marginal density score does not imply hydroclimatic conditional reliability. Reliability deviation follows the same ordering as the interval CRPS, with 0.322 for Global CP, 0.313 for HSCC, and 0.193 for CQR. These metrics show that CQR is competitive on global distributional scoring, while the paper's scientific claim remains specifically about hydroclimatic conditional reliability and the limits of marginal evaluation.

4. Discussion

4.1. Why hydroclimatic boundaries fail

The results point to a specific failure mode: hydroclimatic labels help when they partition calibration residuals within a stable domain, but they do not make residual distributions portable across all regions. In the gauged temporal-split setting, the same basin population supplies calibration and test examples, so tier-specific quantiles can correct the pooled residual mismatch. Under HUC-2 transfer, the tier label must also carry information across regional snowmelt timing, geology, seasonality, and model-error structure. The snow violation shows that this second requirement can fail. Basin-pooled snow coverage falls from 0.873 under Global CP to 0.711 under HSCC (single-seed anchor folds; Section 3.4), even though the HSCC interval is narrower.

Residual variance provides a concise empirical explanation for this behavior. The cross-seed association between calibration-period variance of signed log residuals and test-period signed coverage error (-0.901; residual-dispersion proposition, Section 2.6) indicates that coverage error is strongly ordered by the residual-variance structure, without implying that residual variance is the sole physical cause of miscoverage. It means the conformal score distribution is not invariant across the hydroclimatic and regional cells being compared: when same-tier residual variance differs between source and held-out regions, the tier-specific conformal quantile can become too narrow for the target domain.

4.2. What a diagnostic map buys over a cookbook

The failed cookbook classifier is useful precisely because it prevents over-claiming. A prescriptive cookbook would imply that train-pool attributes are sufficient to predict whether a held-out HUC-tier cell is reliable, marginal, or unreliable. The leave-one-HUC-out test does not support that interpretation: the random forest reaches accuracy 0.344, while the tier-only baseline reaches 0.469, with paired-bootstrap p value 0.926. Thus, reliability is not reliably inferable from the available cell descriptors alone.

The appropriate product is therefore a diagnostic map, not a rulebook. The map records where stratified calibration improves, where it fails, and where dominance checks trigger caution. This is still scientifically useful for PUB-style hydrology, because uncertainty methods are often evaluated by average performance even though scientific interpretation depends on specific basins and regions (Hrachowitz et al., 2013; Pool et al., 2021). The map should be read as scope information for the conditional-miscoverage diagnosis, not as deployment guidance.

4.3. Why this is not a deployment rule

A deployment rule would need reliable ex ante selection among Global CP, HSCC, widened intervals, weighted calibration, or source-pool redesign. The current evidence does not support that level of prescription. The HUC-2 snow violation shows that stratification can under-estimate target-domain residual spread, while the failed cookbook classifier shows that the available cell descriptors do not reliably predict which held-out cells will fail. The static-attribute weighted-conformal rescue attempt also improves mean LORO coverage only by making intervals much wider: mean width rises from 4.05 to 114.97 millimeters per day, a factor of 28.4. This weak rescue is expected because the density-ratio weights use aridity and snow fraction only; they do not reweight the residual-variance features that organize the observed transfer failures. The later fallback-rule audit was also downgraded to exploratory diagnostic status. These results justify warning labels and stress-test reporting, not an operational fallback algorithm.

In the gauged temporal split, HSCC reduces cross-tier spread by 19.44 percentage points, showing that hydroclimatic stratification can reduce the disparity; under spatial transfer the same tier labels become a stress test of exchangeability rather than a solution. Separating these two regimes preserves the conformal validity scope of the method while keeping the manuscript focused on conditional-miscoverage diagnosis.

4.4. Limitations

Several limitations constrain the claims. First, the empirical evidence is CAMELS-centered. CAMELS-US and CAMELS-GB are strong large-sample hydrology benchmarks, but they do not cover all hydroclimatic, management, and data-quality conditions relevant to global streamflow forecasting. A third independent CAMELS-style data set would be needed to test whether the same boundary patterns hold outside the United States and Great Britain.

Second, the multi-seed protocol includes an archived seed-42 run rather than a fully retrained replicate for that seed. The archived run is valuable because it preserves continuity with the original experiment, but it should be described as reuse rather than as an independent retraining event. The manuscript therefore reports cross-seed means as the primary numbers while acknowledging that future replication could rerun that seed under the final software and hardware environment. Relatedly, the HUC-2 leave-one-region-out transfer results, including the headline snow dominance violation, are reported on the seed-42 anchor folds because each fold retrains the backbone from scratch; they are therefore a single-seed transfer estimate without the cross-seed error bars of the gauged five-seed protocol, and a full multi-seed LORO replication is left to future work.

Third, the retained-summary basin-block bootstrap strengthens the basin-sampling argument but remains summary-level. It resamples basins and seeds from retained per-basin summaries and therefore does not test daily temporal dependence or year-block uncertainty; a daily/year-block bootstrap would require regenerating or retaining the non-anchor daily prediction files.

Fourth, the mechanism evidence remains correlational. The controlled synthetic perturbation returned null results for the AR1 pathway and the HSCC coverage pathway, with p values 0.184 and 0.968, respectively. This limits the mechanism claim to residual-variance association and transfer-boundary diagnosis.

Fifth, standard UQ metrics show that split conformal methods optimize marginal coverage but do not control distributional shape. The Kolmogorov–Smirnov (KS) uniform test on the probability integral transform (PIT) gives $p \approx 0$. This is expected because split conformal calibrates only two interval endpoints and the reconstructed predictive density between them is, by construction, uniform rather than a true conditional density, so the PIT cannot be uniform even when interval coverage is on target. The consequence is a scope restriction rather than a contradiction: the reliability studied here is interval coverage by hydroclimatic tier, not full predictive-density calibration, and the tier coverage spreads we report are therefore statements about where the 90 percent interval contains the observation, not about the shape of the predictive distribution. A method with correct tier coverage but a non-uniform PIT can still be misranked by density scores such as CRPS, which is why we report CRPS alongside, rather than in place of, tier coverage. Finally, the HopCPT baseline uses 574 basins rather than the full set of 671 due to strict filtering requirements (see also the cross-family diagnostic comparison in Section 3.7); while the variance of signed log residuals shows no systematic bias in the filtered subset, comparing its tier spread directly to full-set benchmarks requires this caveat.

4.5. A validated calibration-period screening diagnostic for gauged and partially gauged basins

Subject to the threshold-calibration and data-availability limitations stated above, the basin-level variance of signed log residuals is diagnostically informative in three concrete senses. Because it is computed from observed streamflow over the calibration period, it applies to gauged or partially gauged target basins; for truly ungauged basins, where no observed flow is available, the residual-variance proxy must be replaced by attribute-based similarity or model-ensemble disagreement, so the indicator is not a substitute for ungauged-basin prediction.

First, the residual-variance ordering across basins is consistent across hydroclimatic tiers. Computed against the Global-CP signed coverage error of Figure 2, the seed-42 per-tier Spearman of σ_b^2 is -0.66 (dry), -0.73 (semi-arid), -0.90 (humid), and -0.76 (snow); the same ordering persists against the HSCC residual coverage error (-0.76, -0.77, -0.90, -0.82), so stratification does not remove it. The five-seed cross-basin association is -0.90 $\pm$ 0.01, so the diagnostic is neither seed-fragile nor restricted to a single regime.

Second, the diagnostic can flag held-out (region, regime) cells before LORO transfer is applied. We propose the following diagnostic template: (a) compute σ_b^2, the variance of signed log residuals, on the calibration period for each basin in the proposed target pool; (b) compare the target-pool distribution of σ_b^2 to the calibration-pool distribution via Kolmogorov–Smirnov or Wasserstein-1 distance; and (c) if the distribution shift is large, label the resulting stratified intervals as outside the demonstrated exchangeability scope rather than claiming a stratum-specific reliability guarantee. This template is a screening protocol with a clearly bounded scope: it is an actionable audit trigger that flags where stratified intervals leave the demonstrated exchangeability regime, but it is a screen rather than a coverage guarantee.

A threshold-calibrated version of the template makes this warning operational enough for evaluation. In CAMELS-US the out-of-fold basin-level screen ($\sigma_b^2 \geq 0.453$) retains precision 0.74 and recall 0.89 for severe Global-CP undercoverage, and CAMELS-GB requires a different calibrated cutoff and recall-specificity profile (Section 3.3). This is the key limitation and the key message: thresholds should be calibrated within the target basin population or a representative historical pool before being used as audit triggers, and they are screening triggers rather than guarantees.

Third, the diagnostic contrasts with the failed leave-one-HUC-out cookbook classifier (Section 3.4; Section 4.2): rather than predicting which cells will fail, it orders basins by a directly observable proxy, so it is positioned as a calibrated warning indicator rather than a cell-level predictor.

We therefore establish a recommended evaluation protocol, not a single operational threshold. The future-work agenda needed to turn this candidate diagnostic into a deployment rule includes (i) threshold calibration across multiple LSTM architectures and CAMELS-style data sets beyond the United States and Great Britain, (ii) sensitivity to score-function and calibration-window choices, and (iii) integration with multivalid conformal prediction or Gibbs–Candès conditional conformal prediction as alternative downstream reliability targets.

5. Conclusion

This study shows that marginal conformal reliability is not enough for streamflow uncertainty quantification. In CAMELS-US, a pooled split conformal interval leaves large hydroclimatic coverage disparities, with Global CP tier spread 20.97 percentage points. HSCC reduces that spread to 1.53 percentage points in the gauged temporal-split setting, but this result is best read as a mitigation probe for the diagnosis rather than as a universal conditional-reliability solution.

The broader evidence defines the scope of the claim. Under HUC-2 spatial transfer, the snow tier exposes a dominance violation, with single-seed basin-pooled HSCC coverage 0.711, so stratified calibration cannot be presented as a Prediction in Ungauged Basins (PUB) prescription. Vanilla Global CQR, Mondrian-CQR, HopCPT, UMAL, the CAMELS-GB external check, the bounded CAMELS-GB CQR check, Adaptive Conformal Inference, and a static-attribute weighted-conformal rescue attempt all reduce or reshape the same disparity without making marginal coverage a sufficient criterion; the weighted rescue in particular improves LORO spread only with a 28.4-fold width penalty. The contribution is therefore diagnostic and actionable: uncertainty methods for streamflow should be evaluated by where they are reliable, not only by whether their average coverage appears acceptable, and the calibration-period variance of signed log residuals gives practitioners a validated screening signal, computable before deployment, for the gauged or partially gauged basins where marginal coverage is most likely to mislead.

The post-review robustness checks strengthen that diagnostic reading. A basin-block and seed bootstrap preserves the large Global CP versus HSCC spread separation, while a calibration-period residual-variance threshold provides a recommended evaluation protocol for flagging severe undercoverage. Future work should address the remaining limits by testing conditional conformal architectures, such as multivalid conformal prediction or the Gibbs-Candès conditional conformal framework, which formally bound subgroup violations. Extending the hydroclimatic diagnosis to third-continent data sets and evaluating distributional post-processing will help clarify whether distribution-free conditional reliability can be achieved without trading away interval sharpness.

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Author Contributions

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Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

AI Tool Use

AI-assisted coding tools (Claude Code) were used for code prototyping, documentation, and language refinement under author supervision. No AI tool was used to generate data, produce experimental results, or make autonomous scientific interpretations; all AI-assisted code and text were reviewed and verified by the authors, and all experimental design, scientific interpretation, and final text remain the authors' responsibility. No AI tool is listed as an author.

Data Availability Statement

All large-sample data sets used in this work are openly available. The CAMELS-US data set is distributed by the National Center for Atmospheric Research (Addor et al., 2017) and is available at https://doi.org/10.5065/D6MW2F4D. The CAMELS-GB data set is distributed by the UK Centre for Ecology and Hydrology (Coxon et al., 2020) and is available at https://doi.org/10.5285/8344e4f3-d2ea-44f5-8afa-86d2987543a9. The NeuralHydrology library used for LSTM training is open source under the MIT license (Kratzert et al., 2022) and is available at https://github.com/neuralhydrology/neuralhydrology (release v1.13.0).

Processed metrics, conformal-calibration and analysis code, per-basin coverage and width outputs, and figure-generation scripts that reproduce the figures and tables in this manuscript are publicly available on GitHub at https://github.com/QuantDiviner/Reliability-Control-Streamflow. A frozen, reproducible archive of the project code and processed data is deposited on Zenodo under DOI: https://doi.org/10.5281/zenodo.20587508.

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